Tokenisation of Computing Power in Distributed Ledger Technology

Consensus, Verifiable Compute, and Tokenomics under Fiscal Expansion, Grid Scarcity, Memory Limits, and Post-Quantum Transitions

Plauthium is a post-quantum layer-1 protocol for tokenised verifiable computation, specified in the Plauthium Technical Specification [1]. Its native unit, PLAU, has a fixed supply of 15,000,000; chain weight comes only from Fresh Work, an exact integer matrix product, while commercial computation is paid in PLAU and carries no consensus weight.

I. Macroeconomics: Debt and Supply

I.A. Debt and issuance

On 30 September 2026, the close of fiscal year 2026, the total public debt outstanding of the United States reached $40.17 trillion, comprising $32.43 trillion held by the public and $7.75 trillion in intragovernmental accounts (components do not sum to the total because of rounding). Twelve months earlier it stood at $37.64 trillion; at the close of fiscal year 2019, $22.72 trillion; in 2008, $10.02 trillion; and in 2000, $5.67 trillion. Eight days into the new fiscal year the figure reached $40.31 trillion [2]. Between the first fiscal year-ends above $10, $20, $30 and $40 trillion (2008, 2017, 2022 and 2026; Figure 1), the debt grew at compound annual rates of 8.1, 8.9 and 6.8 percent; measured by the daily dates on which each threshold was first crossed (September 2017, January 2022 and 18 August 2026), the last two $10 trillion increments took four years and five months and four years and seven months [2].

On the International Monetary Fund's general-government measure, which consolidates federal, state and local obligations, U.S. gross debt reached 123.9 percent of GDP in 2025 and is projected to reach 142.1 percent by 2031 (Table 1) [3]. Global general-government debt is projected to reach 100 percent of world GDP by 2029, while sovereign interest outlays rose from about 2 percent to nearly 3 percent of global GDP over four years [4]. Sovereign bond issuance across OECD countries is projected at a record $18 trillion in 2026, compared with $12 trillion in 2022 [5].

I.B. Deficits at full employment

Preliminary Congressional Budget Office (CBO) estimates for fiscal year 2026 place federal receipts at $5,403 billion, outlays at $7,396 billion and the federal deficit at $1,993 billion [6]. Net interest on the public debt reached $1,143 billion, or 21.2 percent of federal receipts, exceeding both Medicare outlays net of offsetting receipts ($1,069 billion) and Department of Defense military outlays ($916 billion); only Social Security was larger. Under current law, CBO projects federal debt held by the public to rise from 101 percent of GDP in 2026 to 120 percent in 2036, surpassing the 1946 peak of 106 percent, while baseline net interest rises from 3.3 to 4.6 percent of GDP [7]. IMF staff estimate

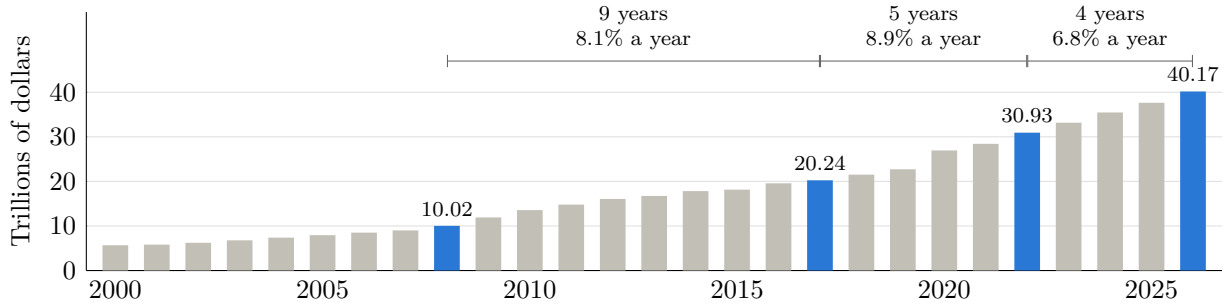


Figure 1: Total public debt outstanding of the United States at fiscal year-end, fiscal years 2000 to 2026, in trillions of nominal dollars. Blue bars mark the first year-end above each multiple of $10 trillion; brackets give the interval between those fiscal year-ends and the compound annual growth rate over it. Source: U.S. Treasury [2].

Jurisdiction / Group	2007	2019	2025	2031
World	59.1	82.1	93.9	102.3
G7	77.8	114.6	123.0	132.8
United States	64.9	108.8	123.9	142.1
Italy	103.5	133.9	137.1	136.1
France	65.5	98.2	116.0	120.7
United Kingdom	43.0	84.9	102.3	102.6
China	28.7	59.8	99.2	126.8
Germany	63.7	58.7	62.9	73.7

Table 1: General-government gross debt, percent of GDP. Values for 2025 are IMF estimates and values for 2031 are IMF projections. Source: IMF, World Economic Outlook database, April 2026 [3].

the U.S. output gap at zero in 2025 and 2026, with the labour market operating at or near full employment [8]. The fiscal imbalance is therefore structural rather than cyclical.

I.C. Monetary aggregates and reserve assets

The Federal Reserve's balance sheet expanded from $0.90 trillion in August 2008 to $4.16 trillion in February 2020 and peaked at $8.97 trillion in April 2022, before quantitative tightening reduced it to about $6.5 trillion by its conclusion on 1 December 2025; reserve-management purchases of Treasury bills begun on 12 December 2025 subsequently raised it to $6.75 trillion as of 7 October 2026 [9, 10]. The M2 money stock rose from $4.67 trillion in January 2000 and $15.49 trillion in February 2020 to $21.79 trillion in March 2022, contracted 4.8 percent by October 2023, and reached $23.34 trillion in August 2026 [9]. Central banks purchased more than 1,000 tonnes of gold, net, in each year from 2022 to 2024 and 863 tonnes in 2025, against an annual average of 473 tonnes from 2010 to 2021 [11].

In a macroeconomic environment defined by structural fiscal deficits and discretionary central-bank balance-sheet policy, Plauthium removes endogenous monetary discretion from its own settlement rules by fixing total supply at Genesis and making every reserve release a deterministic function of chain state [1] (Section V).

II. Energetics: Grid Capacity and Consensus Work

II.A. Electricity demand from computation

A consensus rule built on partial-preimage search over a cryptographic hash function [12] converts electrical power into ledger security on machines that can do nothing else. Bitcoin drew an estimated 138 TWh on an annualised basis in mid-2024, about 0.54 percent of global electricity consumption [13], and approximately 190 TWh in December 2025 according to preliminary estimates from the Cambridge Centre for Alternative Finance [14]. U.S. cryptocurrency mining accounted for 25 to 91 TWh in 2023, or 0.6 to 2.3 percent of national electricity demand [15]. Global data-centre electricity consumption rose from 415 TWh in 2024 [16] to 485 TWh in 2025 (+17 percent), with consumption at AI-focused facilities up 50 percent and total data-centre demand projected by the International Energy Agency (IEA) to reach roughly 950 TWh by 2030 [17]. In the United States, data centres are projected to consume between 6.7 and 12 percent of national electricity by 2028 (Table 2) [18]. The dominant kernel of AI training and inference is dense matrix multiplication.

II.B. Retail prices and wholesale capacity

The average nominal residential electricity price in U.S. cities stood at 13.3 cents per kilowatt-hour in August 2010 and 13.7 cents in August 2020 before rising to 19.6 cents in August 2026, a real increase of 10.5 percent since August 2020 on the CPI electricity index deflated by the CPI-U (Figure 2) [19]. Commercial compute operators contract power through industrial tariffs and power purchase agreements [17], where the binding bottleneck is wholesale firm capacity. PJM Interconnection, serving more than 67 million people across 13 states and the District of Columbia, cleared at the FERC-approved price collar in three consecutive Base Residual Auctions (Figure 3). The 2027/28 auction fell 6,623 MW short of its reliability requirement; in that auction, PJM

Load	Period	TWh per year	Source
Data centres, world	2024	415	[16]
Data centres, world	2025	485	[17]
Data centres, world	2030, projection	about 950	[17]
Data centres, United States	2023	176	[18]
Data centres, United States	2028, projection	325 to 580	[18]
Bitcoin network, world	June 2024, annualised	138	[13]
Bitcoin network, world	Dec. 2025, annualised, preliminary	about 190	[14]
Cryptocurrency mining, United States	2023	25 to 91	[15]

Table 2: Electricity drawn by computation. The source of each figure is cited in its row.

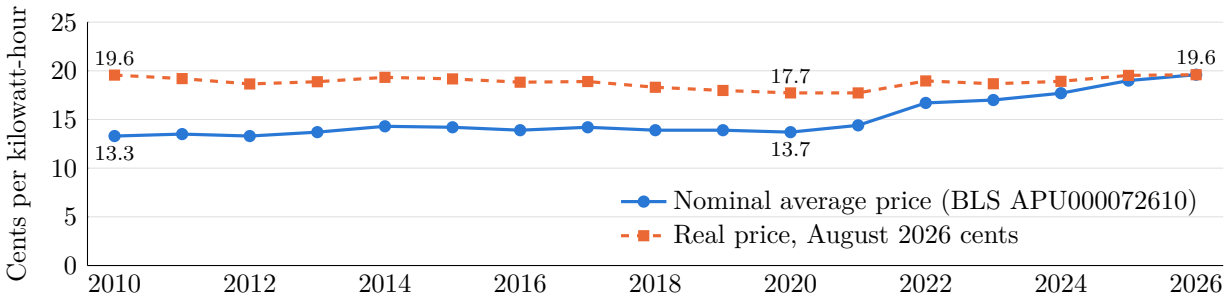


Figure 2: Average residential price of electricity, U.S. city average, each August from 2010 to 2026. The real series applies the change in the CPI electricity index, deflated by the CPI-U, to the August 2026 price. Source: U.S. Bureau of Labor Statistics, series APU000072610, CUUR0000SEHF01 and CPIAUCNS [19].

attributed nearly 5,100 MW of the 5,250 MW increase in forecast peak load to data centres. The 2028/29 auction cleared at the $325.00 per MW-day cap ($554.72 per MW-day uncapped) with a 6,831 MW shortfall [20].

II.C. Dual-use consensus capital and weight isolation

Silicon dedicated to hash preimage search has no use outside the network it secures; industry surveys indicate that the scarce asset held by mining operators is power access [14]. Plauthium instead targets the integer matrix units of general-purpose accelerators: Fresh Work evaluates an exact 512 by 512 INT8 matrix product with INT32 accumulation over matrices expanded from a chain-bound seed [1, §§5.1–5.2], the same algebraic kernel executed by the Scientific Virtual Machine (SciVM) for paid tasks [1, §3.6]. Commercial payloads carry zero consensus weight [1, §1.1]; proof-of-useful-work schemes that use client-chosen matrices [21] let large buyers of computation earn fork-choice weight on work they already fund. Isolating chain weight to seed-bound Fresh Work prevents commercial payloads from conferring fork-choice influence while keeping the underlying accelerators redeployable between consensus and commercial execution. While Fresh Work output has no commercial buyer—its electricity remains a security cost, and the dual use lies in avoiding stranded hardware capital—the seed-bound generator rotates across three deterministic diagnostic profiles (signed outlier arithmetic, nonnegative graph kernels and correlated accumulator stress) that form an open compute-validation benchmark stream [1, §§5.1–5.2].

III. Silicon Concentration and Memory Bounds

III.A. Supply concentration and the memory wall

Over two decades, peak server floating-point throughput grew roughly 60,000-fold while DRAM bandwidth grew 100-fold and interconnect bandwidth 30-fold [22], a divergence known as the memory wall [23]. While large dense matrix multiplications are compute-bound, autoregressive sequence decoding, attention mechanisms [22] and STARK proof generation—which streams low-degree polynomial extensions through Merkle commitment trees [1, §3.5]—are governed by memory capacity and bandwidth.

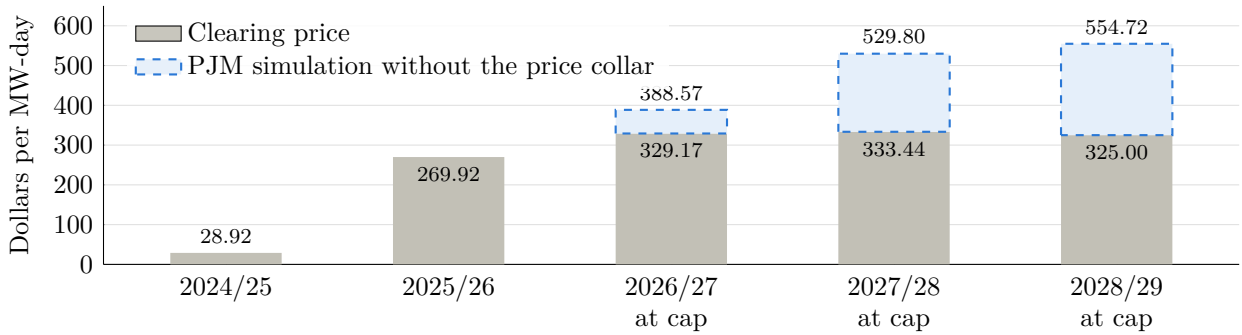


Figure 3: PJM Base Residual Auction clearing prices for the PJM region by delivery year, in dollars per MW-day, with PJM's simulations of the same auctions without the FERC-approved price collar. The 2027/28 and 2028/29 auctions cleared 6,623 MW and 6,831 MW short of the reliability requirement. Source: PJM Interconnection [20].

Layer	Measured concentration	Period
EUV lithography systems	1 manufacturer [25]	2025
Logic capacity below 10 nm	92 percent located in Taiwan, 8 percent in South Korea [27]	2021 study
Foundry revenue	72.5 percent of top-ten foundry revenue at 1 firm [24]	2Q 2026
DRAM revenue	87.6 percent at 3 firms [24]	2Q 2026
AI compute capacity	more than 60 percent on 1 vendor's hardware [28]	2025
Notable AI models	more than 90 percent produced by industry [28]	2025

Table 3: Concentration along the compute stack. The source of each figure is cited in its row; the figure for logic capacity is taken from a 2021 study.

The supply chain behind that bottleneck is concentrated (Table 3). Contract prices for conventional DRAM rose 93 to 98 percent in the first quarter of 2026, and second-quarter DRAM revenue rose 59.5 percent to $154.7 billion, with 87.6 percent captured by three manufacturers [24]. High-bandwidth memory shortages are projected to persist through at least 2027 [17]. A single firm manufactures extreme-ultraviolet lithography systems, recognising EUR 11.6 billion on 48 units in 2025 [25]; a single foundry accounted for $40.9 billion in 2025 capital expenditure [26] and 72.5 percent of top-ten foundry revenue in the second quarter of 2026 [24]. The capital expenditure of five technology firms exceeded $400 billion in 2025 and is projected to rise by a further 75 percent in 2026, surpassing global upstream investment in oil and natural gas production [17].

III.B. Sizing the unit of work and the Ozaki-II profile

So that high-bandwidth memory does not gate proving, Plauthium sizes its canonical work unit to fit commodity GPU memory. Multiplying two 512 by 512 matrices whose INT8 entries lie between -127 and 127 yields exact INT32 entries no larger in magnitude than 8,258,048, embedding without wrap-around into the 61-bit Mersenne prime field over which Plauthium instantiates the Circle STARK [29], [1, §3.1]. A trace of 1,048,576 rows by 128 columns occupies 1 GiB, its two low-degree-extension buffers at a blow-up factor of eight require 16 GiB, and auxiliary buffers take at most 2.5 GiB, bounding peak prover memory at 19.5 GiB—within a single 24 GiB consumer GPU [1, §3.5]. Doubling the trace length would require at least 34 GiB for the trace and its extensions alone, before auxiliary buffers, exceeding the memory of current consumer GPUs. Assumption A1 holds that an attempt on a fresh context costs at least the canonical evaluation, which comprises seed expansion, the matrix product and the full-output commitment; consensus nodes verify winning Fresh Work seals and shares directly in $O(n^2)$ work via six Fiat–Shamir Freivalds rounds without recomputing the product, whereas full Circle STARK proofs are generated for commercial SciVM tasks [1, §§1.5, 3.6, 5.2–5.4].

For commercial and scientific floating-point tasks, the SciVM executes deterministically over the same native 512 by 512 INT8 tiles via a versioned Ozaki-II residue-splitting profile [30]: scaled operands are reduced modulo pairwise-coprime integers no larger than 255 to canonical centred

Milestone or estimate	Value
FIPS 203, FIPS 204 and FIPS 205 published (final) [34]	13 August 2024
Proposed deprecation (NIST IR 8547 ipd): quantum-vulnerable algorithms at 112-bit security, e.g. RSA-2048, ECDSA over P-224 [35]	Deprecated after 2030; disallowed after 2035
Proposed disallowance (NIST IR 8547 ipd): quantum-vulnerable signatures and key establishment at 128-bit security or more, e.g. ECDSA over P-256, EdDSA, RSA-3072 [35]	Disallowed after 2035
Resource bound, 256-bit prime-field ECDLP, Roetteler et al. (2017) [36]	About 2,330 logical qubits; 130 billion Toffoli gates
Resource estimate, secp256k1 ECDLP, Babbush et al. (2026) [37]	At most 1,200 logical qubits, 90 million Toffoli gates; minutes of runtime on under 500,000 physical qubits with planar connectivity at physical error rates of one in a thousand

Table 4: Post-quantum transition milestones and quantum resource estimates. The NIST IR 8547 dates are proposals in an initial public draft (ipd).

residues between -127 and 127, multiplied as exact native tiles, reconstructed via the Chinese Remainder Theorem and canonically rounded with in-circuit error certification [1, §3.6]. NVIDIA's cuBLAS library adopted Ozaki schemes for emulated FP64 matrix multiplication, reaching up to 175 TFLOPS on a B200 GPU [31].

IV. Post-Quantum Transition and Cryptographic Agility

Shor's algorithm solves integer factorisation and discrete logarithms in polynomial time on a fault-tolerant quantum computer [32], rendering RSA, Diffie–Hellman and elliptic-curve cryptography (ECDSA, EdDSA, BLS, Pedersen and KZG commitments) vulnerable. National Security Memorandum 10 sets the U.S. federal goal of mitigating as much of the quantum risk as is feasible by 2035 [33]. Following the publication of finalised standards FIPS 203 (ML-KEM), FIPS 204 (ML-DSA) and FIPS 205 (SLH-DSA) in August 2024 [34], NIST IR 8547 (initial public draft) proposes deprecating 112-bit quantum-vulnerable public-key systems after 2030 and disallowing all quantum-vulnerable signature and key-establishment schemes—including 128-bit elliptic-curve systems such as P-256—after 2035 (Table 4) [35].

ECDSA over secp256k1, which secures Bitcoin and Ethereum, has the same 128-bit classical strength and the same exposure to Shor's algorithm. Estimates of the quantum resources needed to break it have fallen. Whereas Roetteler et al. (2017) bounded 256-bit prime-field elliptic-curve discrete logarithms at roughly 2,330 logical qubits and about 130 billion Toffoli gates [36], Babbush et al. (2026) construct secp256k1 circuits requiring at most 1,200 logical qubits and 90 million Toffoli gates (or 1,450 logical qubits and 70 million Toffoli gates)—the first needing about 1,400 times fewer Toffoli gates than the 2017 bound, with an estimated runtime of minutes on fewer than 500,000 superconducting physical qubits, assuming planar connectivity and physical gate error rates of one in a thousand; the authors withheld the full circuits and published a zero-knowledge proof covering the point-addition subcircuit, patched in version 2 [37]. Any ledger launched on elliptic curves faces an eventual state-wide key migration, and every public key already revealed on chain becomes exposed once such a machine exists; Plauthium avoids this migration by enforcing post-quantum signature and key-establishment schemes and block resource accounting from Genesis.

V. Protocol Architecture and Invariant Mechanisms

Figure 4 illustrates the coupled flows of Fresh Work, commercial jobs and recycled fees in the Plauthium Technical Specification [1], and Table 5 summarises how the protocol responds to the macroeconomic and physical constraints described in Sections I–IV (rows I–IV) and to protocol-internal aggregation and fee-manipulation threats (rows V–VI).

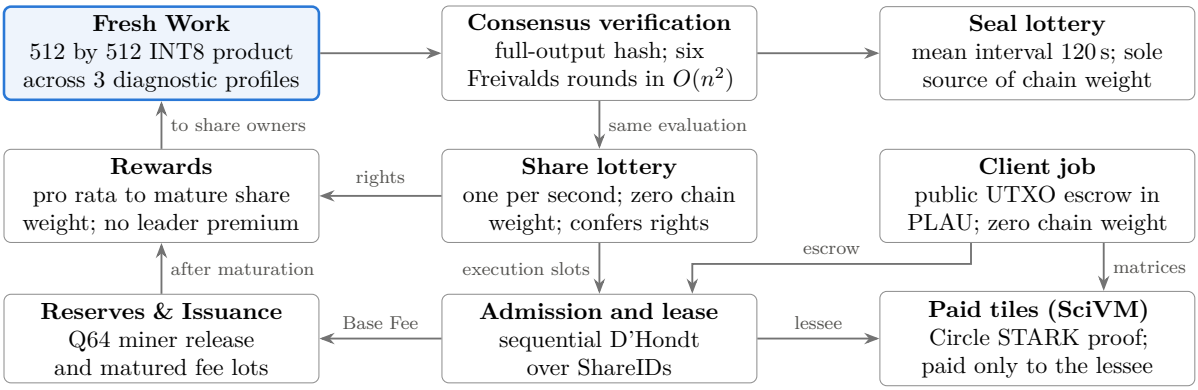


Figure 4: Flows of Fresh Work, client jobs and fees in the Plauthium Technical Specification. Only seals contribute chain weight; commercial payloads, shares and fees contribute none [1, §§1.1, 5.5].

Constraint	**Protocol mechanism and fixed quantities**	**Spec. [1]**
I. Fiscal expansion and monetary discretion (Section I)	Supply is fixed at 1.5 quadrillion atoms (15,000,000 PLAU of 100 million atoms each), conserved across ten disjoint transparent state categories via atomic zero-sum deltas with no shielded pool. All issued balances are zero at Genesis: 90% resides in the unissued miner reserve released via an exact integer Q64 exponential decay (H_M = 1,051,200 blocks), and 10% releases linearly over 1,051,200 blocks into `InitialProtocolPool`, whose gross outflow is capped by a rolling 720-block `InitialValve` at 30,000 PLAU (2%) and requires both an ML-DSA-87 and an SLH-DSA-SHAKE-256s signature.	§§2.2, 7.1, 7.5–7.6
II. Stranded ASIC capital under grid scarcity (Section II)	Fresh Work runs on general-purpose INT8 matrix hardware across three modulo-3 diagnostic profiles that can be redeployed to commercial SciVM tasks, avoiding stranded ASIC capital. In contrast to proof-of-useful-work schemes that use client-chosen matrices [21], Plauthium gives commercial payloads zero chain weight. One evaluation feeds a FruitChains-style [38] dual lottery: seals, every 120 seconds on average, carry weight, and shares, at one per second, confer rights. Under a 2-second delay bound (including $O(n^2)$ Freivalds verification), an adversary holding 10% of Fresh Work reverses a 34-block confirmation with probability below one in a trillion [39].	§§1.1, 3.1, 5.1–5.6
III. Memory wall and access to proving (Section III)	Peak Circle STARK prover memory for a 512 by 512 INT8 tile over the 61-bit Mersenne prime field is at most 19.5 GiB, so any 24 GiB consumer GPU can generate commercial proofs. Six Freivalds rounds [40] verify tiles in 9/256 (about 3.5%) of the multiplication operations, while the algebraic-test soundness of the STARK is set by extension-field challenges and overall profile soundness is bounded in [1, §2.5]. The SciVM executes high-precision floating-point tasks deterministically via Ozaki-II residue splitting and Chinese-remainder reconstruction.	§§2.5, 3.1–3.6
IV. Quantum-vulnerable cryptography (Section IV)	Genesis admits only NIST Category 5 signature and key-establishment schemes: ML-DSA-87 and ML-KEM-1024, the parameter sets specified by CNSA 2.0 [41], with ML-DSA-87 signatures of 4,627 B against 64 B for ECDSA [42]; SLH-DSA-SHAKE-256s as a hash-based co-signature, which CNSA 2.0 does not include; and SHA3-512 for commitments and identifiers. Blocks enforce a five-component resource vector, and a versioned set of active suites selected by block height admits future standards such as FN-DSA [43] without downgrade paths.	§§2.1–2.4, 7.10
V. Capital and identity aggregation (protocol-internal)	The sequential D'Hondt method favours large parties when applied to party totals [44]. Plauthium applies it to discrete, unmergeable ShareIDs whose weights are fixed by the parent block's target, so splitting or merging wallets earns no premium, and payment only to the executor named in a tile's lease removes execution races and priority auctions.	§§6.1–6.7
VI. Fee manipulation and self-dealing (protocol-internal)	Base Fees move at admission to the locked reserve and are never refunded to the payer. On maturity they are released as independent integer fee lots, distributed pro rata to mature share weight alongside primary Q64 miner issuance, with rounding dust permanently segregated. Self-dealing loses money in expectation whenever the coalition's discounted, fairness-adjusted payout share is below one.	§§7.7–7.9

Table 5: Macroeconomic and physical constraints (rows I–IV) and protocol-internal threats (rows V–VI) mapped to the rules and fixed quantities of the Plauthium Technical Specification [1].

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